

GeoVision: Designing a Complementary Human-AI Workflow for Cue-Based Geolocation

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Image geolocation, the process of determining the precise location on Earth where a photo or video was created, is a fundamental task for Open-Source Intelligence (OSINT) analysts in many domains, including investigative journalism, law enforcement, and historical scholarship. We present GeoVision, a complementary human-AI workflow in which AI surfaces cues, retrieves guide knowledge, and ranks candidates, while humans reconcile conflicting evidence and make precise final judgments.

1 INTRODUCTION AND RELATED WORK

Image geolocation determines where on Earth an image was captured, often based on manual analysis of visual cues such as signage, road markings, architecture, infrastructure, and vegetation. It is an open-source intelligence (OSINT) technique foundational to investigative journalism, law enforcement, and historical scholarship [1,2,9]. Geolocation requires extreme accuracy, as an exact location can provide new leads, establish actionable search radii, and determine an accurate record of historical events [1,2].

Large-language models allow for potentially semantically accessible queries while geolocating, but findings indicate participants struggle to properly communicate needs in a concise and guided manner [10].

Vision-language models (VLMs) have increased dramatically in performance and can quickly identify features, retrieve information, and propose locations, but often are biased toward guessing large cities[4–8], struggle in rural cue-sparse areas [3,9], and are vulnerable to adversarial image manipulation [5]. Models have improved performance by implementing retrieval-augmented generation (RAG) by vectorizing existing community guides of unique geographic features [6]. State-of-the-art models achieve around 60% city-level (25km) and 27% street-level (1km) accuracy [5,8]. This level of granularity is often insufficient and still requires humans for further location refinement and verification. This efficient but somewhat inaccurate nature of VLMs offers a unique complementary workflow opportunity.

Our system, GeoVision, therefore proposes a complementary human-AI workflow where VLMs detect cues, retrieve knowledge, and rank candidates, while users inspect evidence, steer attention, and make the final call.

2 GEOVISION DESIGN PROCESS

Across three 45-minute sessions, four novice participants completed rounds of *OpenGuessr* [12], an online geolocation game where players must determine location from panoramic street view images. They also annotated images, evaluated existing resources, and tested prototype VLM/RAG tools. We analyzed observations and feedback for recurring breakdowns, usability issues, and autonomy preferences.

3 GEOVISION PROTOTYPES AND ITERATIVE TESTING

The first GeoVision prototype we tested was a Streamlit [11] application into which users uploaded street view screenshots. The prototype identified scene features and candidate countries with explanations and confidence (Figure 1a). The Claude-Sonnet-4-6 VLM supported cue detection, guide retrieval, and candidate generation, while users retained responsibility for interpreting evidence and making the final judgment. The findings from post-play semi-structured group interviews highlighted that participants value connections between visible cues and guide

knowledge, as well as VLM identification of subtle cues that might go unnoticed. However, participants also mentioned that the screenshot upload workflow was cumbersome and disrupted the geolocation process, and that outputs displayed high confidence toward incorrect answers. Participants desired transparent links between VLM-identified cues and predictions, and a more interactive model to direct the VLM toward specific scene elements.

These observations informed the feedback-driven concept in Figure 1b, an overlay street view plugin that supports an iterative loop of inspecting cues, comparing evidence, steering model attention, and revising hypotheses. Additionally, the system indexes geographic observational information from the Plonk It guide [13], converting it into embeddings for semantic retrieval, and presents users with categorical drop-down lists and explanations. This enables users to identify cues the VLM might have missed or incorrectly identified, and maintain autonomy during the geolocation process.

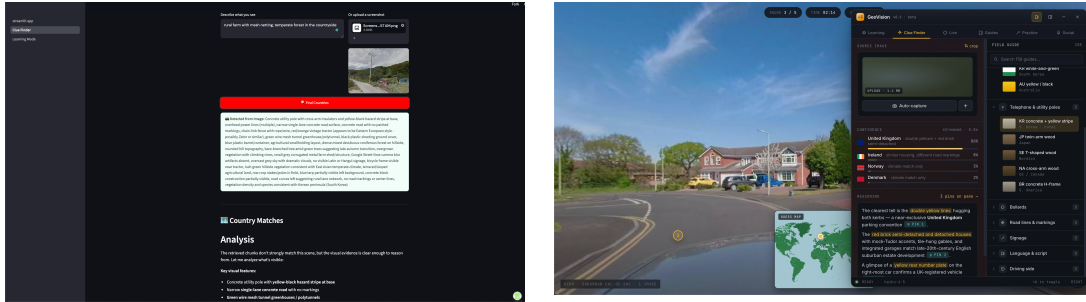


Figure 1: GeoVision evolved from (a) a separate interactive chatbot prototype to (b) an integrated overlay with ranked hypotheses, uncertainty, evidence links, and user-steerable attention.

4 PRELIMINARY DESIGN INSIGHTS

From this user testing and development process, we propose four design implications for future human-AI geolocation systems that we plan to validate through formal evaluation.

(1) Preserve human control over evidence. Expose the cues behind each location candidate so users can inspect, challenge, and combine them before choosing a location. **(2) Retrieve knowledge from observed features:** connect a selected pole, road marking, sign, or building feature directly to relevant guide examples without requiring an initial country hypothesis. **(3) Communicate uncertainty at the supported scale:** rank alternatives and distinguish regional evidence from country or street-level evidence to reduce overreliance on precise but weak predictions. **(4) Make model attention steerable through intuitive, human-AI interaction:** allow users to identify relevant cues and guide model attention through spatial annotations, such as pins and bounding boxes, reducing the need to verbally articulate complex visual cues.

5 FUTURE WORK

The next step is to implement the integrated overlay and evaluate whether this human-AI workflow improves performance over existing VLMs and/or manual human geolocation. A controlled experiment will compare AI-only, human-AI, and human-only conditions on distance error and time. This data will allow for analyzing confidence calibration, resolution of conflicting cues, and density-level differences. Logs and interviews will test whether feature selection and evidence-linked explanations help users correct model errors and steer the final stages of geolocation.

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